Google unveils TensorStore, a library to help manage large data sets

The open source C++ and Python framework TensorStore, developed by Google, aims to accelerate the design for reading and writing huge multi-dimensional arrays. Multi-dimensional data sets that cover a single large coordinate system are commonly used in contemporary computer science and machine learning applications. These data sets are challenging to work with because customers frequently wish to conduct investigations involving numerous workstations operating in parallel, and may receive and output data at unpredictable intervals and varied scales.

Google Research has developed TensorStore, a library that provides users with access to an API that can manage huge data sets without the requirement for sophisticated hardware, to address issues with data storage and manipulation. Numerous storage systems, including local and network file systems, Google Cloud Storage, and others are supported by this library.

To load and work with enormous arrays of data, TensorStore provides a simple Python API. Any arbitrary big underlying data set can be loaded and updated without having to store it in memory, because no actual data is read or kept in memory until the precise slice is required.

This is made possible by the indexing and manipulation grammar, which is quite similar to the syntax used for NumPy operations. TensorStore supports virtual views, broadcasting, alignment, and other sophisticated indexing features like data type conversion, down sampling, and haphazardly created arrays.

Additionally, it includes an asynchronous API that enables read or write operations to work concurrently.

Legit Security announces new tool to secure GitHub configurations

Legitify, an open source security tool to secure GitHub implementations, has been announced by Legit Security, a cyber security company with an enterprise platform to secure an organisation's software supply chain. Security teams and

FIWARE Foundation and Red Hat join hands to develop a smart city solution

Red Hat, Inc., has announced a partnership with the FIWARE Foundation, a non-profit organisation that promotes the use of open standards in the creation of smart solutions, to create an integrated platform that will use data to make cities all over the world more resilient and improve the wellbeing of their residents. Red Hat Open Innovation Labs collaborated with the FIWARE Foundation and Human Oriented Products (HOPU), a solution provider involved in the FIWARE community, over the course of a six-week residency to develop an enhanced smart city solution that is simple to deploy, fully scalable, reliable and runs on Red Hat OpenShift.

Cities around the world are dealing with escalating and complicated environmental, economic, and social issues that have an impact on the daily lives of their residents. City officials and public sector decision-makers need eco-smart technology solutions to solve these issues so they can use data to assist them make decisions that are good for the environment and for their constituents.

The smart city platform powered by FIWARE is able to derive insights from data processed by HOPU air quality sensors in order to adopt wiser decisions for the welfare of inhabitants. It can be deployed on any private or public cloud. This platform and its base application for air quality monitoring have been implemented in Las Rozas and Cartagena, two Spanish cities, as pilot use cases.